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E1.31 (DMX) / Art-Net

E1.31

WLED supports the E1.31 (sACN) realtime light protocol.

> **Version Info**
>
> **As of WLED v0.10.0, Art-Net is alternatively supported.** All information on this page applies for Art-Net as well. You will need to switch to Art-Net mode in Sync settings and reboot once.

> **Version Info**
>
> **As of WLED v0.11.0, DDP is alternatively supported.** Using DDP, the Multi RGB DMX mode is always used regardless of the DMX mode setting (as it is no DMX) You will need to switch to DDP mode in Sync settings and reboot once.

> **Version Info**
>
> **As > WLED 0.14.0-b1, DMX Effect mode channel mapping changed.** This is a breaking change in E131 sync behavior. Existing DMX setups using WLED Effect mode are likely to break. Solution: Adopt external DMX channel mappings according to the new Effect mode channel layout.

Features

- 170 LEDs (510 DMX channels) are supported per universe.
- You can use up to 9 adjacent universes which is a total of more than the maximum 1500 LEDs WLED can drive.
- E1.31 priority handling. HTP for multiple senders with same priority is not implemented!

Note: WLED > 0.14.0-b1 implements a basic priority handling. To enable it set E1.31 port priority UI config > 0. Packages < config priority will always be skipped. Higher priority senders will overrule lower priority package senders (timeout 3 seconds). Non-zero start code and E1.31 preview data is ignored.

Note: If you see issued with lag, you may see better performance if you disable Wi-Fi Multimedia (WMM) Mode QoS on your router

General Recommendations

- For a fluent 25ms (40fps) or more framerate, it is recommended to use no more than 3 universes (510 LEDs).
- Using multiple E1.31 sources to control a single WLED controller will impact performance. If possible it's recommended to avoid this.
- Configure WLED to connect to a WiFi Network. It was observed that sending E1.31 from a computer to WLED when accessed as access point (WLED-AP) may not work, see forum post.
- When using an ESP32, disabling WiFi sleep in the WiFi settings of WLED can reduce/eliminate visual stuttering/lag of effects, and significantly reduce ping times. Note: disabling WiFi sleep will increase power consumption.

Using LedFX

We highly recommend the LedFx project in combination with WLED! Configure WLED first, ensure devices are powered on and connected to your WiFi. To add WLED devices, in the 'Device Management', click on the Find WLED devices button. If you want to manually add devices, use more than 170 LEDs with LedFx, you need to go to additional settings and set "Universe Size" to 510 when adding the WLED device!

Settings

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DMX types

Select the mode you want to use in Sync settings.

Disabled

Incoming E1.31 packets will be ignored.

Single RGB

All LEDs are set to the same color. 3 Channels: Red, Green, Blue

Single DRGB

All LEDs are set to the same color. 4 Channels: Master Dimmer, Red, Green, Blue

Effect

Not a realtime mode & only support 1 universe. Allows setting WLED effect properties over E1.31 with 15 channels.

Channel	Property
1	Master Dimmer
2	Effect mode ID
3	Effect speed
4	Effect intensity
5	Effect palette ID
6	Effect option
7	Red Primary
8	Green Primary
9	Blue Primary
10	Red Secondary
11	Green Secondary
12	Blue Secondary
13	Red Tertiary
14	Green Tertiary
15	Blue Tertiary

The `effect option` channel is divided into 128 macro parts to control the various states that a segment can be in.

Using 2D effect options on a 1D strip has no effect.

Value	Reverse	Mirror	1D to 2D Map (2D only)	Transpose (2D only)	Mirror Y (2D only)	Reverse Y (2D only)
0..1			Pixels			
2..3			Pixels			x
4..5			Pixels		x	
6..7			Pixels		x	x
8..9			Pixels	x		
10..11			Pixels	x		x
12..13			Pixels	x	x	
14..15			Pixels	x	x	x
16..17			Bar			
18..19			Bar			x
20..21			Bar		x	
22..23			Bar		x	x
24..25			Bar	x		
26..27			Bar	x		x
28..29			Bar	x	x	
30..31			Bar	x	x	x
32..33			Arc			
34..35			Arc			x
36..37			Arc		x	
38..39			Arc		x	x
40..41			Arc	x		
42..43			Arc	x		x
44..45			Arc	x	x	
46..47			Arc	x	x	x
48..49			Corner			
50..51			Corner			x
52..53			Corner		x	
54..55			Corner		x	x
56..57			Corner	x		
58..59			Corner	x		x
60..61			Corner	x	x	
62..63			Corner	x	x	x
64..65	x		Pixels			
66..67	x		Pixels			x
68..69	x		Pixels		x	
70..71	x		Pixels		x	x
72..73	x		Pixels	x		
74..75	x		Pixels	x		x

76..77	x			Pixels	x	x	
78..79	x			Pixels	x	x	x
80..81	x			Bar			
82..83	x			Bar			x
84..85	x			Bar		x	
86..87	x			Bar		x	x
88..89	x			Bar	x		
90..91	x			Bar	x		x
92..93	x			Bar	x	x	
94..95	x			Bar	x	x	x
96..97	x			Arc			
98..99	x			Arc			x
100..101	x			Arc		x	
102..103	x			Arc		x	x
104..105	x			Arc	x		
106..107	x			Arc	x		x
108..109	x			Arc	x	x	
110..111	x			Arc	x	x	x
112..113	x			Corner			
114..115	x			Corner			x
116..117	x			Corner		x	
118..119	x			Corner		x	x
120..121	x			Corner	x		
122..123	x			Corner	x		x
124..125	x			Corner	x	x	
126..127	x			Corner	x	x	x
128..129		x		Pixels			
130..131		x		Pixels			x
132..133		x		Pixels		x	
134..135		x		Pixels		x	x
136..137		x		Pixels	x		
138..139		x		Pixels	x		x
140..141		x		Pixels	x	x	
142..143		x		Pixels	x	x	x
144..145		x		Bar			
146..147		x		Bar			x
148..149		x		Bar		x	
150..151		x		Bar		x	x
152..153		x		Bar	x		
154..155		x		Bar	x		x

156..157		x	Bar	x	x	
158..159		x	Bar	x	x	x
160..161		x	Arc			
162..163		x	Arc			x
164..165		x	Arc		x	
166..167		x	Arc		x	x
168..169		x	Arc	x		
170..171		x	Arc	x		x
172..173		x	Arc	x	x	
174..175		x	Arc	x	x	x
176..177		x	Corner			
178..179		x	Corner			x
180..181		x	Corner		x	
182..183		x	Corner		x	x
184..185		x	Corner	x		
186..187		x	Corner	x		x
188..189		x	Corner	x	x	
190..191		x	Corner	x	x	x
192..193	x	x	Pixels			
194..195	x	x	Pixels			x
196..197	x	x	Pixels		x	
198..199	x	x	Pixels		x	x
200..201	x	x	Pixels	x		
202..203	x	x	Pixels	x		x
204..205	x	x	Pixels	x	x	
206..207	x	x	Pixels	x	x	x
208..209	x	x	Bar			
210..211	x	x	Bar			x
212..213	x	x	Bar		x	
214..215	x	x	Bar		x	x
216..217	x	x	Bar	x		
218..219	x	x	Bar	x		x
220..221	x	x	Bar	x	x	
222..223	x	x	Bar	x	x	x
224..225	x	x	Arc			
226..227	x	x	Arc			x
228..229	x	x	Arc		x	
230..231	x	x	Arc		x	x
232..233	x	x	Arc	x		

234..235	x	x	Arc	x		x
236..237	x	x	Arc	x	x	
238..239	x	x	Arc	x	x	x
240..241	x	x	Corner			
242..243	x	x	Corner			x
244..245	x	x	Corner		x	
246..247	x	x	Corner		x	x
248..249	x	x	Corner	x		
250..251	x	x	Corner	x		x
252..253	x	x	Corner	x	x	
254..255	x	x	Corner	x	x	x

Effect + White

Same as `Effect` with additional whites = 18 channels.

Channel	Property
..	..
16	White Primary
17	White Secondary
18	White Tertiary

Effect Segment

Same as `Effect` with 15 channels per segment; expect `channel 1 = "Segment Dimmer"`.

All effect segment modes introduce an additional `DMX segment spacing`. If spacing `s = 0` subsequent DMX addresses for all segments are created. When `s > 0` a gap of `s` DMX addresses between segments is used. To calculate segment DMX fixture addresses:

```
Segment DMX Address (i) = DMXAddress + i * (dmxEffectChannels + s)
```

Where `DMXAddress` = start address as configured in UI, `dmxEffectChannels` = 15|18 depending on selected effect segment mode, `s` = address gap as configured in UI and `i` is the index id of each segment as existing.

Note: 1 DMX universe = max 512 DMX addresses. So max number of segments depends on start address, selected segment mode and configured DMX segment spacing. To calculate how many segments can be controlled:

```
Max segments = floor[ (512 - DMXAddress) / (dmxEffectChannels + s) ]
```


Same as `Effect Segment` including whites, so it uses 18 DMX channels per segment.

Multiple RGB

3 Channels per LED in sequence. LED 0 Red, LED 0 Green, LED 0 Blue, LED 1 Red, ... Default mode, equivalent to pre-0.9.1 E1.31. This is the mode you want to use for xLights and LedFx.

Multiple RGBW

Like `Multiple RGB` + additional white channels.

Dimmer + Multi RGB

Like `Multiple RGB`, but the first channel is a brightness control. Master Dimmer, LED 0 Red, LED 0 Green, ...

Preset

Trigger presets and control brightness. 2 Channels: Brightness, Preset ID